BOOKLEAF PUBLISHING

Technical Assignment

Full-Stack Developer Position

Assignment 1 of 2

Document: Author Support & Communication Portal
Duration: 4 days from date of receipt
Deliverables: Source code, live demo, documentation:

This document contains all the information, data, and context you need to complete the assignment. Please read it thoroughly before you begin.

CONTEXT

BookLeaf Publishing is a self-publishing company that processes 1,200+ books monthly across a catalog of 22,000+ titles. We work with thousands of authors across India and the US.

One of our biggest operational challenges is author communication at scale. Authors frequently reach out with questions about their royalties, book status, ISBN issues, and printing quality. Currently, these queries are handled manually, leading to delayed responses, inconsistent communication, and a growing backlog—especially during high-volume periods.

Your assignment is to build a solution to this problem.

THE ASSIGNMENT

Build an **Author Support & Communication Portal** — a web application with two interfaces (author-facing and admin-facing) powered by AI-assisted response generation. The system should help BookLeaf's team handle author support queries faster, more consistently, and with less manual effort.

> ⚠️ Important: We are not looking for pixel-perfect UI. We are evaluating your architecture, code quality, AI integration, and how you think about real-world product problems. A clean, functional interface is sufficient.

DETAILED REQUIREMENTS

A. Author Portal (Author-Facing)

A simple dashboard where an author can log in and interact with their BookLeaf account.

- **Authentication:** Email-based login. No need for OTP or social login—simple email/password is fine. Each author should only see their own data.
- **My Books:** A page showing the author's published books with key details: title, ISBN, genre, publication date, status, MRP, total copies sold, total royalty earned, royalty paid, and royalty pending.
- **Submit a Support Query:** A form where the author can raise a ticket. The form should include: a dropdown to select which book the query is about (or "General / Account Level"), a subject line, a detailed description field, and optional file attachment support (UI only—actual upload is bonus, not required).
- **My Tickets:** A page showing all tickets the author has submitted, with current status (Open, In Progress, Resolved, Closed), the original query, and any responses from the admin team. Tickets should update in real time or near-real time—if an admin responds, the author should see it without refreshing the page.

B. Admin Portal (Admin-Facing)

An internal dashboard for the BookLeaf operations team to manage incoming support queries.

- **Ticket Queue:** A list of all incoming tickets across all authors, filterable by status, category, priority, and date. The most urgent and oldest unresolved tickets should be easy to spot.
- **AI-Powered Auto-Classification:** When a new ticket comes in, the system should automatically classify it into one of these categories using an AI model: Royalty & Payments, ISBN & Metadata Issues, Printing & Quality, Distribution & Availability, Book Status & Production Updates, or General Inquiry. The admin should be able to override the AI's classification if it's wrong.
- **AI-Generated Priority Score:** Each ticket should get an automatic priority score (Critical / High / Medium / Low) based on the content. For example, a "I haven't received any royalty for 6 months" should be flagged higher than "Can I update my author bio?" The admin should be able to override this too.
- **AI-Drafted Response:** When an admin opens a ticket, the system should present an AI-generated draft response based on the query content AND the BookLeaf Knowledge Base provided below. The admin should be able to edit the draft before sending. The sent response should appear on the author's ticket view.
- **Ticket Management:** Admins should be able to update ticket status, add internal notes (not visible to the author), and assign tickets to themselves.

C. API Layer

The frontend and backend should communicate through a clean REST API. We expect:

- Properly structured endpoints with consistent naming conventions.
- Authentication middleware protecting all routes.
- Role-based access control: authors can only access their own data; admins can access everything.
- Input validation and meaningful error responses (not generic 500s).
- API documentation—either a Swagger/OpenAPI spec, a Postman collection, or a well-structured README section explaining every endpoint.

D. AI Integration Details

This is a critical part of the evaluation. We want to see how you think about integrating AI into a production application, not just that you can make an API call.

1. **Prompt Engineering:** Your prompts should be well-structured, use the Knowledge Base context we've provided, and produce responses that sound like a real BookLeaf support representative—not generic AI fluff.
2. **API Key Security:** API keys must be stored in environment variables, never hardcoded or exposed to the frontend. We will check for this.
3. **Error Handling:** What happens when the AI API is down or rate-limited? The system should degrade gracefully—tickets should still be created, and admins should be able to write responses manually.
4. **Cost Awareness:** Are you sending the entire ticket history + knowledge base on every call, or are you being smart about token usage? We want to see evidence that you've thought about this.

5. **Model Choice:** You may use OpenAI, Anthropic, or any other LLM API you prefer. Justify your choice briefly in your documentation.

BOOKLEAF KNOWLEDGE BASE

Provide the following information to your AI system as context for generating responses. This is how BookLeaf actually operates—your AI-drafted responses should reflect these policies accurately.

Company Overview

- BookLeaf Publishing is a self-publishing company operating in India and the US.
- We offer publishing packages: Standard Free (no upfront cost) and Bestseller Breakthrough (premium, paid package with marketing and distribution add-ons).
- We handle cover design, typesetting, ISBN assignment, printing, distribution, and royalty management for our authors.
- Our in-house printing facility and warehouse are located in Delhi. We also work with print partners including Repro India and Epitome Books.

Royalty Policy

- BookLeaf follows an 80/20 royalty split: 80% of the net profit per book goes to the author, 20% to BookLeaf.
- Net profit = MRP minus printing cost, platform commission (Amazon/Flipkart), and shipping charges.
- Royalties are calculated quarterly and paid within 45 days of the quarter ending.
- Minimum payout threshold: ₹1,000. If accumulated royalties are below this, they roll over to the next quarter.
- Payouts are made via bank transfer to the account linked in the author's dashboard.
- Authors can view a detailed royalty breakdown in their dashboard, showing sales figures per platform.

ISBN Policy

- Every book published through BookLeaf receives a unique ISBN assigned by BookLeaf.
- ISBNs are registered under BookLeaf's publisher imprint. If an author wants an ISBN under their own imprint, they need to obtain it independently.
- If an author reports an ISBN error (duplicate, wrong book linked), it is treated as a high-priority issue and escalated to the production team.

Printing & Quality

- In-house printing handles most orders. Overflow or specific format requirements go to Repro India or Epitome Books.
- Standard print turnaround: 5–7 business days from order confirmation.
- If an author reports a quality issue (misprints, binding defects, color inconsistency), BookLeaf arranges a free reprint after verification. The author may need to share photos of the defective copy.

Distribution & Availability

- Books are listed on Amazon India, Flipkart, Amazon US, Amazon UK, and the BookLeaf Store.
- New listings typically go live within 7–10 business days after publication is complete.
- If a book is showing as unavailable on a platform, it usually indicates a stock sync issue—BookLeaf's team can trigger a re-sync within 24–48 hours.

Production Stages

- A book goes through the following stages: Manuscript Received → Editing (if opted) → Cover Design → Typesetting → Proofreading → ISBN Assignment → Printing → Distribution Setup → Published & Live.
- Authors are updated at each stage via email. Delays typically happen at Cover Design (waiting for author approval) and Proofreading (revision rounds).

Communication Tone Guidelines

This is how BookLeaf communicates with authors. The AI-generated responses should match this tone:

- Always empathetic and professional. Authors are our partners, not customers to be managed.
- Acknowledge the author's concern before jumping to solutions.
- Be specific: include actual numbers, dates, and statuses wherever possible rather than vague reassurances.
- If something is BookLeaf's fault (delayed royalties, ISBN error), own it directly. No corporate deflection.
- If the issue requires escalation or investigation, give a clear timeline ("Our team will look into this and get back to you within 48 hours") rather than open-ended promises.
- Always end with a clear next step for the author and/or the BookLeaf team.

Sample Query → Response Examples

Below are examples of the kind of queries BookLeaf receives and how responses should be framed. Use these to calibrate your AI's output.

Category	Sample Author Query	Expected Response Tone & Content
Royalty & Payments	I published my book 4 months ago and still haven't received any royalty. What's going on?	Acknowledge the frustration. Explain the quarterly royalty cycle and 45-day payout window. Check if their bank details are linked. Provide the specific next payout date. If the payment is genuinely overdue, escalate with a 48-hour resolution timeline.
Royalty & Payments	My royalty amount seems too low. I sold 200 copies but only received ₹3,000.	Explain the net profit calculation: MRP minus printing, platform commission, and shipping. Offer to share a detailed line-by-line royalty breakdown. Don't be defensive—offer transparency.
ISBN & Metadata	My book is showing a different ISBN on Amazon than what's on the physical copy.	Treat as high priority. Acknowledge this is a serious data issue. Confirm you are escalating to the

		production team immediately. Provide a 48-hour resolution timeline.
Printing & Quality	I received my author copies and the print quality is terrible. The images are blurry and pages are misaligned.	Apologise sincerely. Ask the author to share photos of the defective copies. Confirm that BookLeaf will arrange a free reprint once verified. Provide a timeline of 5–7 business days for the reprint.
Distribution	My book is published but it's showing as "Currently Unavailable" on Amazon.	Explain that this is typically a stock sync issue. Confirm you're triggering a re-sync with the distribution team. Set expectation: 24–48 hours for it to go live again.
Production Updates	It's been 3 weeks and my book is still in typesetting. When will it be done?	Check the actual production status. If it's delayed, be honest about why (e.g., waiting on author approval for proofs). Provide a specific updated timeline. Don't blame the author even if the delay is on their side—frame it collaboratively.
General	Can I update the description of my book on Amazon after it's already live?	Yes, metadata updates can be submitted through the dashboard or by emailing the BookLeaf team. Changes typically reflect on platforms within 3–5 business days.

SAMPLE DATASET

A JSON file (bookleaf_sample_data.json) is provided alongside this document. It contains 10 authors and 18 books with realistic data including publication statuses, royalty figures, sales numbers, and distribution details.

Use this data to seed your application. Your system should work with this data out of the box—we will test using these exact records.

> Note: The dataset includes books in various states—some published and earning royalties, some with pending payouts, some with zero payouts, and two still in production. This variety is intentional. Your system should handle all these edge cases gracefully.

TECHNOLOGY GUIDELINES

- **Frontend:** React, Next.js, Vue, or any modern JavaScript framework. Your choice—pick what you're strongest in.
- **Backend:** Node.js (Express/Fastify), Python (FastAPI/Django), or any backend framework. Again, your choice.
- **Database:** PostgreSQL, MongoDB, or any database you prefer. We care about your schema design, not which database you use.
- **AI/LLM:** OpenAI, Anthropic, or any LLM API. Use your own API key for development. If cost is a concern, OpenAI's gpt-4o-mini or Anthropic's Claude Haiku are inexpensive options that work well for this use case.
- **Real-Time:** WebSockets, Server-Sent Events, or polling—your choice. The ticket updates just need to feel responsive.
- **Deployment:** Deploy the application somewhere accessible (Vercel, Railway, Render, a VPS—anything). We need a live URL to test.

WHAT TO SUBMIT

1. **Source Code:** A public GitHub/GitLab repository. Add the following email as a collaborator: [we will provide this]. Clean commit history showing your progression is a plus.
2. **Live Demo:** A working, deployed URL where we can test both the author and admin portals. Pre-seed it with the provided dataset. Include test login credentials in your README.
3. **README Documentation:** This should cover: how to set up and run the project locally, your architecture decisions and why you made them, how the AI integration works (prompt strategy, error handling, cost management), API documentation or a link to it, and any known limitations or things you'd improve given more time.
4. **Brief Write-Up (max 1 page):** A short note explaining your approach: what you prioritised, what trade-offs you made, and how you'd evolve this into a production system if given the opportunity.

EVALUATION CRITERIA

Here is exactly how your submission will be scored. We are sharing this upfront so there are no surprises.

Criteria	Weight	What We're Looking For
AI Integration Quality	25%	Prompt design, knowledge base usage, response quality, classification accuracy, graceful degradation, cost-consciousness. This is the highest-weighted criterion.
Architecture & Code Quality	25%	Clean project structure, separation of concerns, meaningful abstractions, readable code, proper error handling. No spaghetti.
API Design	15%	RESTful conventions, authentication, role-based access, input validation, error responses, documentation.
Functionality & Completeness	15%	Does everything work? Both portals, ticket lifecycle, real-time updates, edge cases (empty states, books in production, zero royalties).
Product Thinking	10%	Does the app feel like something a real author and a real ops team would use? Thoughtful UX decisions, sensible defaults, practical workflows.
Documentation & Communication	10%	README quality, architecture explanation, write-up clarity, commit messages, inline code comments where needed.

TIMELINE & PROCESS

You have **4 days** from the date you receive this assignment. We understand you may have a current job—the timeline accounts for that. If you need an extension for a legitimate reason, just let us know before the deadline; we're reasonable about it.

After submission, if your assignment clears our evaluation, we will schedule a 45–60 minute technical discussion where you'll walk us through your code, architecture decisions, and AI integration approach. This is a conversation, not an interrogation—we want to understand how you think.

IMPORTANT NOTES

.

- You are free to use AI tools (Copilot, ChatGPT, etc.) during development. We use AI extensively ourselves. But you must be able to explain every line of your code in the follow-up discussion.
- If anything in this document is unclear, email us. Asking good questions is a positive signal, not a negative one.
- We value working software over perfect software. If you're running short on time, prioritise the core ticket flow and AI integration over nice-to-have features.

Good luck. We're genuinely excited to see what you build.

Team BookLeaf